2.6 Feasibility Study

Technical feasibility is supported by the implemented operating limits, not only by tool availability. The dashboard requests aircraft data every 1 s and status every 4 s. Live aircraft are removed after 45 s without a frame; replay records use a 12 s stale limit. A trajectory point is stored when it is the first sample, at least 5 s has elapsed, movement is at least 30 m, altitude changes by 20 m, heading changes by 5 degrees, or speed changes by 2 m/s. Prediction uses the latest 4-20 samples and is limited to feed gaps of at most 180 s; if the linear fit is unavailable, constant-speed/heading kinematics is used. SQLite runs locally in WAL mode, so no separate database server is required and a 4 GB system is sufficient for the project workload. The main technical limits are map-tile/network availability, receiver data quality, and reduced accuracy during sharp turns or long outages. The system is suitable for academic monitoring, not certified air-traffic control.